Brain Tumor Classification Using CNNs

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Abstract

Image classification and analysis are crucial tasks in numerous fields of medicine. Specifically, the detection and classification of brain tumors can lead to correct diagnoses and treatments. We have designed two customized Convolutional Neu- ral Networks (CNN) to classify brain MRI scans into four categories: glioma tumor, pituitary tumor, meningioma tumor, or no tumor. Glioma tumors originate in the glial cells of the brain and spinal cord. Pituitary tumors are located in the pituitary gland at the base of the brain. Meningioma tumors arise from the meninges, which are protective membranes that cover the brain. This work advances progress in data science on large medical imaging datasets. The same model architecture can be used to perform other medical image classification tasks.

Keywords: CNN, Brain tumor, Machine learning, Keyword4

1 Introduction

Brain tumors stand as one of the leading causes of cancer-related death in the mod- ern world, posing significant challenges in early detection and effective treatment. [1] They can manifest in various regions, often remaining asymptomatic until later stages

of life. As a result, symptoms such as personality changes, memory loss, communication impairments, hearing or speech challenges, chronic migraines, and vision loss frequently go undetected until the disease has progressed considerably. [1]

The application of deep learning in medical image classification has emerged as a transformative approach in medical diagnostics and treatment planning. Convolutional neural networks (CNNs) have shown a particularly significant promise in improving the accuracy and efficiency of diagnosing various medical conditions from imaging data. In recent years, several studies have demonstrated the potential of CNNs in medical image classification. For example, advancements were documented by Li et al. (2014), who classified lung image patches with interstitial lung disease with a customized CNN [2]. Cai et al. (2020), who investigated the application of deep learning models to classify brain tumors [3]. Their study illustrated the effectiveness of CNNs in distinguishing between different types of brain tumors, thus supporting the potential of deep learning in enhancing neuro-oncology diagnostics.

Srinivasan et al. used a CNN, trained and tested on the BraTS2019 dataset, to classify brain tumors into five types: normal, glioma, meningioma, pituitary, and metastatic, which achieved 93.81 percent accuracy [1]. In addition, there have been efforts to apply deep learning for semantic segmentation, one such method is enabling UNet CNNs to detect and extract specific features. For example, the segmentation analysis of different brain regions for early detection of Alzheimer's and Parkinson's disease by Cai et al. (2020) [3]. Sarraf et al. (2017) used the LeNet-5 CNN framework to train Alzheimer's disease samples for sMRI and fMRI, achieving 98.84 percent and 96.85 percent accuracy [4]. This research highlights the continuous improvement in CNN architectures and training methodologies, contributing to the increasing accuracy and reliability of medical image classification and segmenta- tion systems. Furthermore, Kim et al. (2022) explored the efficacy of transfer learning,

a method that involves knowledge transfer between related tasks to improve performance on a new task [5]. This method has made great contributions to the field of medical image analysis by overcoming problems such as data, time, and hardware scarcity. The work of Wang et al. (2019) underscored the critical role of large annotated data sets and fine-tuning methods in training robust and accurate CNN models for medical image classification [6]. Their comprehensive review of advances in the field emphasized the importance of data quality and diversity in achieving high diagnostic precision, particularly in complex tasks such as the classification of liver lesions. Other research in the field of deep learning in image classification has also been tested. Deep neural networks, or DNNs, are artificial neural networks with multiple hidden layers, making them more complex and resource-intensive than conventional neural networks. Seetha and Raja achieved an accuracy rate of 96.97 and 97.5 percent in the analysis of DNN and CNN based brain tumor classification, respectively [7].

The integration of deep learning into clinical workflows has been explored by Litjens et al. (2017) [8]. They discussed the challenges associated with the deployment of deep learning models in clinical settings, including issues related to model interpretability, regulatory compliance, and integration with current workflows. Their insights offer an overview of the practical considerations and the potential future adoption of deep learning in medical settings. In general, these studies collectively demonstrate significant advances in the application of deep learning for medical image classification. The continuous evolution of CNN architectures, coupled with the availability of large annotated datasets, has paved the way for more accurate and efficient diagnostic tools. As this field progresses, it holds great promise in improving patient outcomes through early and precise detection of various medical conditions. We address the limitations of previous studies by comparing them. The

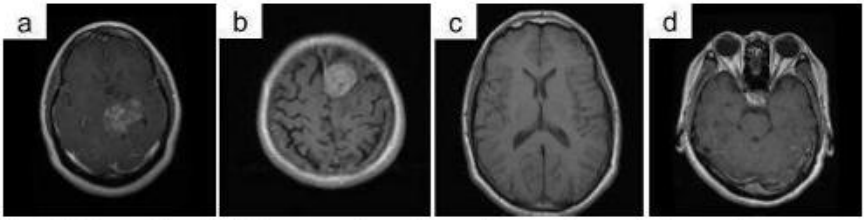


Fig. 1: Example of MRI scans of **a.** glioma tumor, **b.** meningioma tumor, **c.** no tumor, and **d.** pituitary tumor

primary objective of this study is to investigate the application of deep learning techniques in the classification of brain tumor MRI images. Specifically, this study aims to develop, train, and assess the practicality of two different Convolutional Neural Net- works (CNNs) for brain tumor classification. This will include a standard CNN without pretraining and a transfer learning model using a pre-trained EfficientNetB0 base from ImageNet.

2 Results

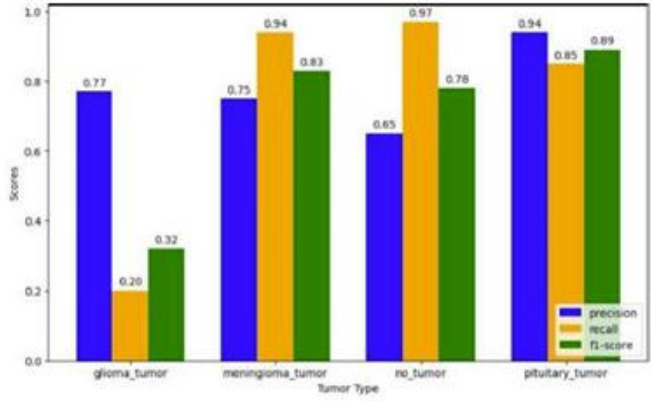


Fig. 2: Evaluation metrics values for each tumor prediction for CNN with pretrained weights

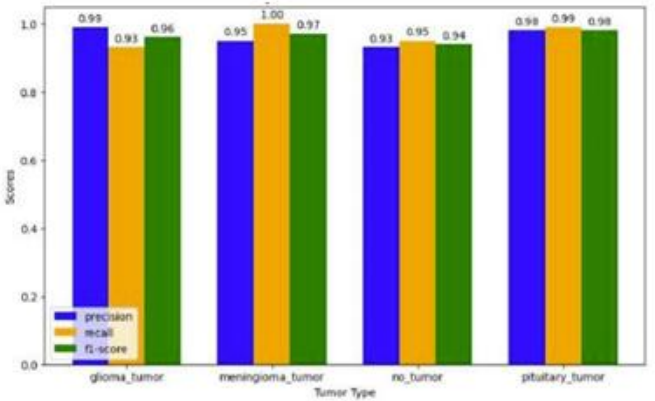


Fig. 3: Evaluation metrics values for each tumor prediction for CNN with pretrained weights

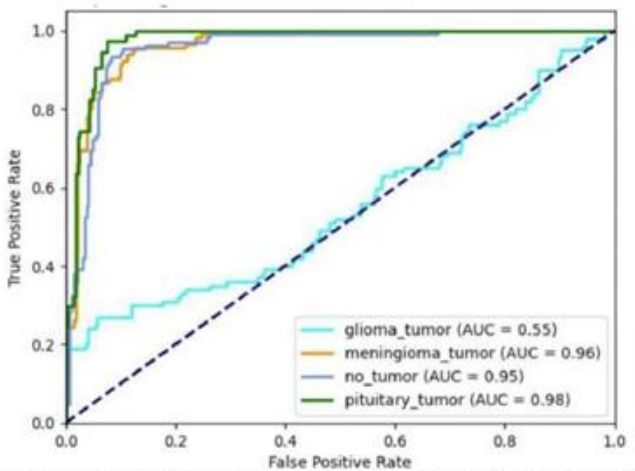


Fig. 4: ROC curves and AUC values of each tumor prediction for CNN without pre-trained weights

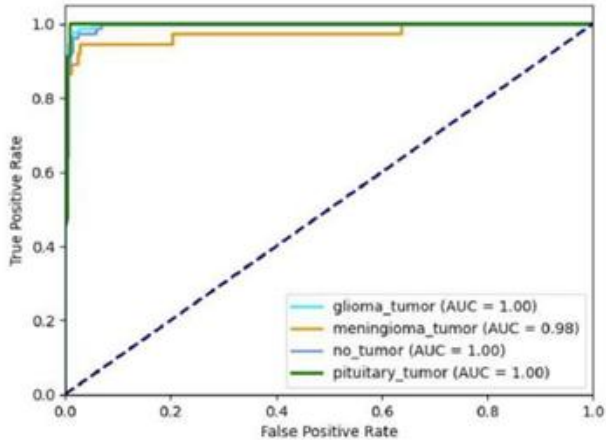


Fig. 5: ROC curves and AUC values of each tumor prediction for CNN with pretrained weights

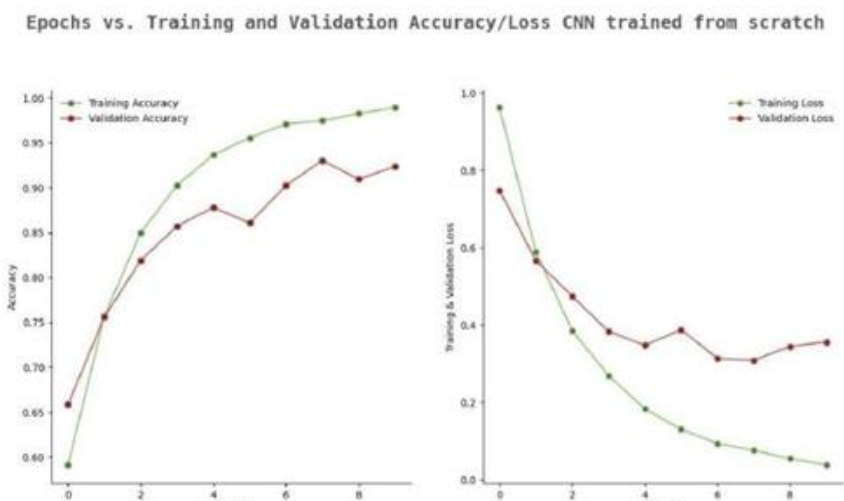


Fig. 6: Training and Validation Accuracy vs Loss (CNN without pre training)

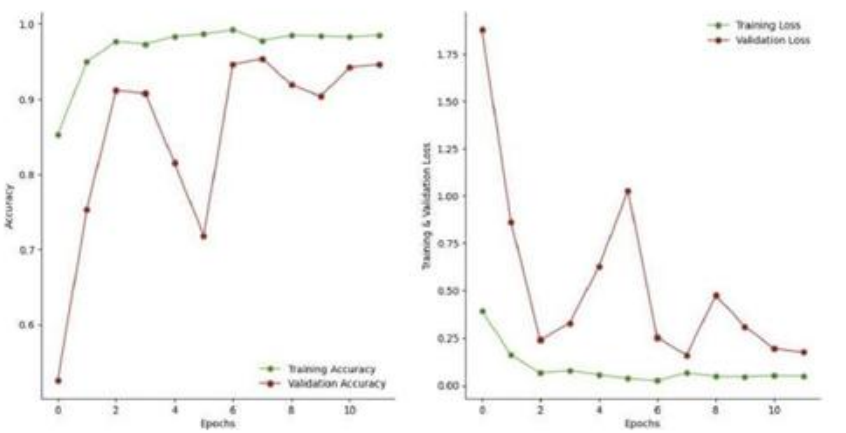


Fig. 7: Training and Validation Accuracy vs Loss (CNN with Efficientnetb0 weights)

2.1 Interpretation of Evaluation Metrics

Overall, the CNN without pre training demonstrated a limited capability to . While it achieved a training accuracy of 98.8 percent across 10 epochs with a loss of 0.05, its performance on the test set was significantly lower, with an accuracy of 73.9 percent, with a loss of 3.3. This discrepancy between the training and test accuracies and high loss value suggests that the model may have overfitted to the training data. This could have been due to many factors such as simpler methods for data aug- mentation and a high dropout value. For the other evaluation metrics, the precision ranged from 0.62 to 0.92, its recall ranged from 0.17 to 0.97, the F1-scores ranged from 0.28 to 0.81. The large variation highlights the limitations of a simpler model, the recall scores indicative of an increased number of false negative predictions. In con- trast, the CNN with the EfficientNetB0 Base achieved higher accuracy, both during training and when tested on unseen data. During

training, the first model achieved an

accuracy of over 97.6 percent across 12 epochs with a loss of 0.17. When evaluated on the test set, the accuracy was slightly lower at 94.1 percent with a loss of 0.22. In addition, the precision, recall, and F1-scores were between 0.85 to 0.96 for all tumor types, indicating that the model performed well across all the classes, with minimal overfitting issues. The consistent performance across all metrics indicates that this model is not only accurate but also reliable in distinguishing between different tumor types and identifying the absence of tumors.

2.2 Interpretation of ROC curve and AUC values

The ROC curve is a graphical representation of a model's ability to distinguish between classes. It plots the true positive rate, also known as sensitivity (The propor- tion of actual positive cases that were predicted correctly) against the false positive rate, or 1-specificity (The proportion of actual negative cases that were predicted incorrectly). The area under the ROC curve (AUC) is a key metric used to summa- rize the model's performance. An AUC of 1.0 indicates perfect classification, while an AUC of 0.5 suggests no discrimination ability (equivalent to random guessing). For the CNN without pre training, the ROC curve showed varying degrees of perfor- mance across the different tumor types. Most notably, the model had an AUC score of 0.55 on the glioma class, compared to the other classes which ranged from the mid to high 0.9s. There was a significantly lower value in the glioma tumor's sensi- tivity/recall (0.17), which had drastically impacted the ROC curve's true positive rate. Several small factors could have contributed to this result, such as the slightly lower data availability in the test set or the model's tendency to overfit on that particular class. While there isn't an apparent "main factor" that caused this, it was likely that the combination of many underlying causes played a role in leading the model to per- form poorly in predicting the glioma class on the testing dataset. The CNN with the EfficientNet base, on the other hand, had a much larger area under the ROC curve,

scoring near perfect AUC values across all tumor classes. This suggests that transfer learning, particularly with a complex model pre-trained on large-scale datasets like ImageNet, can provide a substantial boost in performance for complex medical image classification tasks.

2.3 Interpretation of Training and Validation Loss and Accuracy

2.3.1 Training

Both the CNN without pre training and the EfficientNet base CNN experienced their training accuracy steadily increasing over the epochs, eventually reaching around 98 percent and 97 percent respectively. This indicates that the models are learning the training data well, improving its ability to correctly classify the training samples. Both models' training losses also show a consistent decrease, which aligns with the increasing accuracy, confirming that the model is fitting the training data more closely over time.

2.3.2 Validation

The CNN without pre training also had its validation accuracy increasing, but at a slower rate compared to the training accuracy, leveling off at around 90 percent. This suggests that while the model is learning, it might not be generalizing as well to the validation data as it is fitting the training data. This is credited to the 90-10 training validation split in data, showing how more data is always helpful for models to output correct predictions. Similarly, the Effnet base CNN The validation loss decreases initially but then stabilizes and even increases slightly towards the end, which is a classic sign of overfitting. This means that the model is starting to memorize the training data rather than learning to generalize to new, unseen data.

3 Discussion

3.1 Clinical Implications

The high accuracy and consistency of the EfficientNet-based CNN suggest its poten- tial as a valuable tool in clinical settings for brain tumor diagnosis. Its reliable performance could aid in the medical field, making more accurate and timely diagno- sis, ultimately improving patient outcomes by ensuring that the correct treatment is administered based on accurate tumor classification.

3.2 Limitations of the Study

This study's primary limitation is the overfitting observed in the CNN without pre train- ing, suggesting a need for more sophisticated regularization techniques to prevent overfitting or a more extensive dataset. Additionally, the reliance on a relatively small dataset from Kaggle may limit the generalizability of the findings to broader clinical applications.

3.3 Future Work

Future work should focus on mitigating overfitting in CNNs without pre-training by incorporating advanced regularization techniques or expanding the dataset. Explor- ing additional transfer learning approaches and ensemble methods could further enhance model accuracy and robustness.

3.4 Conclusion

In summary, while the CNN with the EfficientNet base is robust and generalizes well to new data, the CNN without pre training struggles with generalization, particu- larly in the glioma tumor type. This comparison underscores the importance of using methods of transfer learning, pre-trained models, such as EfficientNetB0, which can leverage learned features from large datasets to improve performance in challenging

tasks like brain tumor classification. Applying these methods to other types of medical imaging and conditions could extend the applicability of deep learning in healthcare diagnostics.

4 Methods

4.1 Data Collection

Table 1: Data overview of the training and testing sets

	Testing Set	Training Set
Number of Glioma Tumor Scans	100	826
Number of Meningioma Tumor Scans	115	822
Number of No Tumor Scans	106	395
Number of Pituitary Tumor Scans	75	827

The data set used for classification was sourced from Kaggle, consisting of 3264 images labeled for brain tumor classification. This was split into sets for train- ing, consisting of 2870 images, and testing, consisting of 394 images. The images were classified into pituitary tumor, meningioma tumor, glioma tumor, and no tumor. (**Figure 1**).

4.2 Preprocessing Techniques

Pre-processing refers to the series of steps taken to transform the data into a suitable format for modeling. In the context of brain image classification, preprocessing typically includes tasks such as resizing images to ensure uniform input dimensions, normalizing pixel values to a standard range, and applying data augmentation techniques to artificially increase the diversity of the training dataset. Ensuring con- sistent input formats steps is crucial for the accuracy of the model. In both CNNs, the following preprocessing steps were taken.

4.2.1 Image Resizing

All images were resized to a uniform size. 150x150 pixels for the CNN without pre-training and 224x224 pixels for the EfficientNet model. This ensures consistency in input dimensions for neural networks.

4.2.2 Normalization

The pixel values of the images were normalized to a range of [0, 1] by dividing each pixel value by 255. This process is important for faster convergence during training and helps prevent vanishing gradients.

4.2.3 Data Augmentaton

To enhance the diversity of the training dataset and to avoid overfitting, data augmen- tation techniques were applied. These included random rotations, width and height shifts, horizontal flipping, and zooming. These transformations ensured that the model is more generalized by learning from various altered versions of the images.

4.2.4 Categorical Encoding

The labels of the tumor classes were converted to a binary matrix format. Using the technique of one-hot encoding, each label was turned into a row of numbers with a single "1" to indicate the correct category and "0s" elsewhere. This step was essential for training the model using categorical cross-entropy as the loss function.

4.2.5 Train-Test Split

The data set was pre-divided into a training set and a testing set, which was done by Kaggle. The training set was used to train the model, while the testing set was reserved for evaluating the model's performance on unseen data (1).

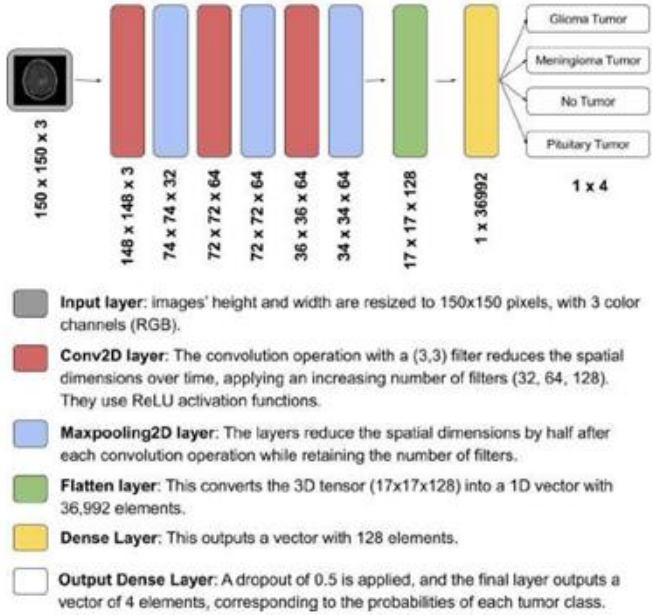


Fig. 8: A visual representation of the architecture from the CNN without pretraining.

References

[1] Srinivasan, S., Francis, D., Mathivanan, S.K., Rajadurai, H., Shivahare, B.D., Shah, M.A.: A hybrid deep cnn model for brain tumor image multi- classification. BMC Medical Imaging **24**(1), 21 (2024) https://doi.org/10.1186/s12880-024-01195-7

[2] Li, Q., Cai, W., Wang, X., Zhou, Y., Feng, D.D., Chen, M.: Medical image classifi- cation with convolutional neural network. In: 2014 13th International Conference on Control Automation Robotics Vision (ICARCV), pp. 844–848 (2014). https://doi.org/10.1109/ICARCV.2014.7064414

[3] Cai, L., Gao, J., Zhao, D.: A review of the application of deep learning in medical image classification and segmentation. Ann Transl Med **8**(11), 713 (2020)

[4] Sarraf, S., DeSouza, D.D., Anderson, J., Tofighi, G., Alzheimer's Disease Neuroimaging Initiativ: Deepad: Alzheimer's disease classification via deep convolutional neural networks using mri and fmri. bioRxiv (2017) https://doi.org/10.1101/070441 https://www.biorxiv.org/content/early/2017/01/14/070441.full.pdf

[5] Kim, H.E., Cosa-Linan, A., Santhanam, N., Jannesari, M., Maros, M.E., Ganslandt, T.: Transfer learning for medical image classification: a literature review. BMC Med Imaging **22**(1), 69 (2022)

[6] Wang, W., Liang, D., Chen, Q., Iwamoto, Y., Han, X.-H., Zhang, Q., Hu, H., Lin, L., Chen, Y.-W.: In: Chen, Y.-W., Jain, L.C. (eds.) Medical Image Classification Using Deep Learning, pp. 33–51. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-32606-7_3 . https://doi.org/10.1007/978-3-030-32606-7_3

[7] Seetha, J., Raja, S.S.: Brain tumor classification using convolutional neural networks. Biomedical and Pharmacology Journal **11**(3), 1457–1461 (2018) https://doi.org/10.13005/bpj/1511

[8] Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., van der Laak, J.A.W.M., van Ginneken, B., Sa´nchez, C.I.: A survey on deep learning in medical image analysis. Medical Image Analysis **42**, 60–88 (2017) https://doi. org/10.1016/j.media.2017.07.005